Factoring out the greatest common factor (GCF)

1

a 3 **b** a **c** $3x$ **d** $3b$

2

a $4r + 12 = 4\,(r + 3)$ **b** $5v - 30 = 5\,(v - 6)$

c $18c - 24 = 6\,(3c - 4)$ **d** $2t^2 + 2t = 2t\,(t + 1)$

3

a $6\,(v + 5)$ **b** $5\,(9t - 8)$ **c** $-2\,(s + 5)$ **d** $-2\,(6s - 5)$

e $y\,(y + 4)$ **f** $k\,(k - 5)$ **g** $2u\,(u - 4)$ **h** $2t\,(2 + t)$

i $x\,(42 - x)$ **j** $7\,(x + 5)$ **k** $6\,(6 - x)$ **l** $-2\,(x - 5)$

m $-10\,(10 + x)$ **n** $-6\left(3y^2 - 1\right)$ **o** $13\,(3x + 4)$ **p** $(t - 3)\,(8t + 9)$

Additional questions

4 When factoring $5xy$ out of the term $-5xy$ you get $\dfrac{-5xy}{5xy} = -1$ but Xander elimintated the term instead.

5

a Let the width of the smallest field be x so that its area is x^2. The largest field has an area of $9x^2$ and the middle field has an area of $x(x + 5)$.

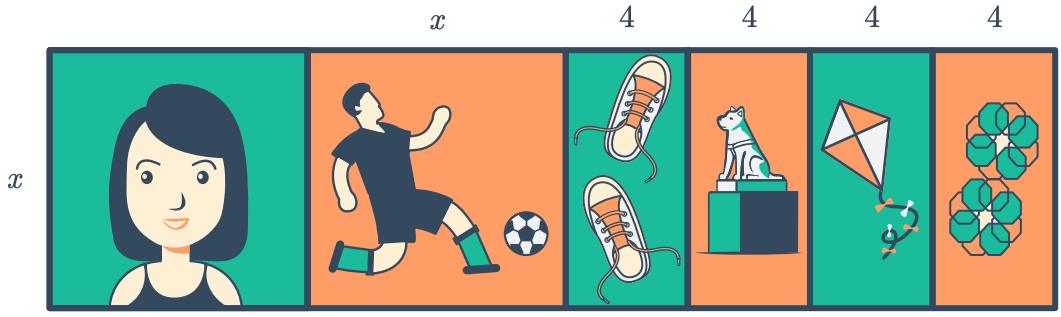
b The area of the entire field is the sum of each smaller field: $x^2 + 9x^2 + x(x + 5)$. The factored form of this expression is $x(11x + 5)$ so the width of the field is x and the length is $11x + 5$

6 When the greatest common factor is a monomial, we can use monomial division to determine what expression is left after the factoring is complete.

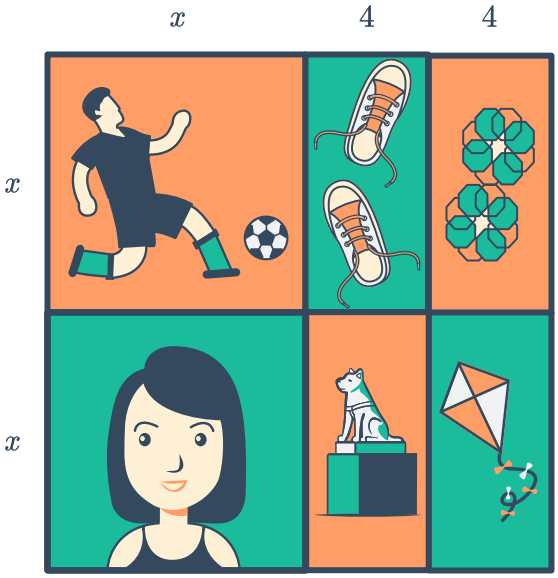
7 The greatest common factor represents the largest value that can be divided evenly from each term. If the greatest common factor of a variable had an exponent larger than any of the original terms, then dividing it out would leave a negative exponent.
For example, consider the expression $x^2 + x^3$. If we divide both terms by x^2 (the smaller exponent) then we are left with $1 + x$. If instead we try to divide both terms by x^3 (the larger exponent) then we would be left with $x^{-1} + 1$.

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a $x(2x + 16)$



b $2x(x + 8)$



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